Project Charter

Embeddable Cross-Course AI Agent General Architecture Platform

1. Project Identification

Field	Details
Project Name	Embeddable Cross-Course AI Agent General Architecture Platform
Date of Authorization	March 29, 2026
Project Start Date	March 30, 2026
Project End Date	June 14, 2026
Version	1.0

2. Project Stakeholders

Name	Title / Role	Student ID	E-mail
冯俊财	AI Architect / Backend Engineer	2353924	20614667@qq.com
路清怡	Backend Engineer	2352996	849014041@qq.com
张诗蔻	Frontend Engineer	2353240	2987991635@qq.com
纪鹏	Test & Documentation Engineer	2351869	jp3274313334@163.com

3. Project Description

Background:
The current higher education sector suffers from "fragmented intelligent teaching." Each course independently develops its own AI teaching support system, leading to architectural heterogeneity, duplicate construction, and high integration costs. Existing systems lack a unified standard for

embedding into mainstream Learning Management Systems (LMS) such as Chaoxing (超星) and DingTalk (钉钉). Furthermore, course-specific knowledge cannot be transferred or reused across disciplines, and assignment grading remains generic—lacking fine-grained annotation feedback that simulates a teacher's hand-written corrections.

Description of the challenge or opportunity:
There is a pressing need for a standardized, reusable AI Agent architecture that enables different courses to rapidly build and deploy intelligent teaching assistants without redundant engineering effort. Large Language Model (LLM) technology now makes it feasible to provide annotated, context-aware feedback at scale—transforming AI grading from "usable" to "precisely useful."

Overview of the desired impact:
This project delivers **EduAI**, a general-purpose AI Agent platform that: (1) provides a standardized SDK for rapid course-Agent development; (2) enables seamless embedding into Chaoxing and DingTalk via a lightweight iframe Widget; (3) performs annotation-style assignment grading with ≥ 95% accuracy; (4) analyzes student learning states and generates targeted exercises to close individual knowledge gaps.

4. Measurable Organizational Value (MOV)

The MOV defines the measurable value this project must deliver to justify its investment.

> **Primary MOV:** Enable universities to reduce the cost and time of building intelligent teaching systems for new courses, while delivering grading feedback quality that matches or exceeds manual teacher annotation.

Area	Goal	Metric	Target Value
Quality	Deliver accurate in-course Q&A	Answer accuracy on core knowledge points	≥ 95%
Quality	Deliver teacher-level grading feedback	Key error annotation accuracy (location + content)	≥ 95%
Cost	Reduce duplicate AI system development effort	Reuse one SDK for ≥ 2 course types without code changes	Demonstrated in demo
Performance	Support large-scale concurrent access	Concurrent users supported	≥ 500

Area	Goal	Metric	Target Value
Adoption	Enable embedding in mainstream LMS platforms	Platforms supported via iframe/Widget	≥ 2 (Chaoxing, DingTalk)

5. Project Scope

What is included in the scope of this project:

- Unified AI Agent SDK (`AgentBase` , `QAAgent` , `GradingAgent` , `ExerciseAgent`) supporting multiple course types
- Knowledge middleware: document ingestion (PDF/Word/PPT), vector embedding (pgvector), knowledge graph, hybrid retrieval (vector + BM25 + Rerank)
- Refined intelligent engine: annotation-style grading (text, code, image), learning analytics (mastery matrix, early warning), personalized exercise generation
- RESTful API backend (FastAPI, 8 route groups: Auth, Courses, Agents, Assignments, Chat, Analytics, Exercises, Platform)
- Full frontend management portal (React 18, 9 pages): Login, Dashboard, Course Management, Intelligent Q&A, Assignment Management, Grading Result Viewer, Analytics Dashboard, Exercise Center, Agent Visual Builder
- Embeddable Widget (iframe-compatible lightweight chat/grading window for LMS integration)
- Platform adaptation layer: technical specification and simulated interfaces for Chaoxing (LTI) and DingTalk (H5)
- Cloud deployment: Docker Compose + Nginx + HTTPS on a cloud server

What is outside the scope of this project:

- Native mobile application (iOS/Android)
- Real-time live-class features (video streaming, screen sharing)
- Full production API certification and live integration with Chaoxing or DingTalk (simulated adaptation only, as permitted by competition rules)
- Automated billing or payment systems
- Support for physical lab equipment or hardware-linked experiments

6. Project Schedule Summary

- **Project start date:** March 30, 2026

- **Project end date:** June 14, 2026
- **Duration:** 11 weeks

Key Milestones:

Milestone	Description	Target Date
M1	Environment running; LLM API connected; database tables created; JWT auth working	April 12, 2026
M2	Q&A MVP: upload course material → AI answers from course knowledge (accuracy ≥ 90%)	April 26, 2026
M3	Full teaching loop: submit assignment → AI annotates → learning analytics → exercises generated	May 17, 2026
M4	Platform embedding + Agent visual builder complete; Q&A ≥ 95%, grading ≥ 95%	May 31, 2026
M5	System fully deployed on cloud server; end-to-end demo validated; project documentation complete	June 14, 2026

7. Requirement Breakdown Structure (RBS)

1.0 Functional Requirements

1.1 Unified AI Agent Framework
Support rapid development and standardized deployment of course-specific AI Agents across different disciplines.

1.2 Knowledge Middleware
Standardized representation, storage, and retrieval of course knowledge; support cross-course knowledge transfer and reuse.

1.3 Platform Adaptation
Provide standardized interfaces for embedding into Chaoxing and DingTalk via iframe / H5 Widget.

1.4 Intelligent Q&A
Support multi-turn course Q&A; answer accuracy on core knowledge points ≥ 95%.

1.5 Assignment Grading & Annotation
Support multi-modal assignment grading (text, code, image); provide inline annotation feedback at error locations, simulating teacher hand-written corrections; key error annotation accuracy ≥ 95% (location + content).

1.6 Learning Analytics & Early Warning
Perform multi-dimensional analysis of individual and class-level knowledge weaknesses based on grading results and learning behavior; provide early warning for at-risk students.

1.7 Personalized Exercise Generation
Automatically generate or match targeted exercises based on identified knowledge weaknesses; implement "assess–grade–practice" closed loop.

1.8 Agent Visual Builder
Provide a visual, drag-and-drop tool for teachers to construct and customize course AI Agents without deep technical knowledge.

2.0 Non-Functional Requirements

2.1 Performance: support ≥ 500 concurrent users
2.2 Accuracy: Q&A ≥ 95%; grading annotation ≥ 95%
2.3 Architecture: microservice-based, independently deployable components
2.4 Security: role-based access control (teacher / student)
2.5 Course coverage: support multiple course types via the same SDK

3.0 Constraints

3.1 Primary LLM: Tongyi Qianwen (qwen-max / qwen-vl-max)
3.2 Platform integration: technical specification only; real Chaoxing / DingTalk certification not required
3.3 Timeline: fixed end date June 14, 2026

8. Work Breakdown Structure (WBS) & Hours Estimation

1.0 Project Management 24 hrs
1.1 Project initiation & charter 4 hrs

1.2 Weekly progress meetings & tracking 12 hrs
1.3 Risk monitoring & issue resolution 4 hrs
1.4 Final review & submission coordination 4 hrs

2.0 Requirements & System Design 40 hrs
2.1 Requirements elicitation & RBS 8 hrs
2.2 System architecture design 12 hrs
2.3 Database schema design 8 hrs
2.4 API interface specification 8 hrs
2.5 UI/UX wireframes (9 pages) 4 hrs

3.0 Infrastructure & DevOps 32 hrs
3.1 Docker Compose environment setup 8 hrs
3.2 Database migration (Alembic) 4 hrs
3.3 Async task configuration (Celery + Redis) 8 hrs
3.4 File storage setup (MinIO) 4 hrs
3.5 Cloud server deployment (Nginx) 8 hrs

4.0 Backend Development 160 hrs
4.1 Core infrastructure (auth, database models, seed data) 24 hrs
4.2 Agent SDK (AgentBase, QAAgent, GradingAgent, ExerciseAgent, LLM provider integration) 40 hrs
4.3 Knowledge middleware (document parsing, embedding, vector storage, knowledge graph) 28 hrs
4.4 Refined intelligent engine (grading, multimodal, learning analytics, exercise generation) 40 hrs
4.5 API routes (8 groups) 28 hrs

5.0 Frontend Development 96 hrs
5.1 Project setup & shared components 8 hrs
5.2 Login / Register page 4 hrs
5.3 Dashboard 8 hrs
5.4 Course management page 10 hrs
5.5 Intelligent Q&A chat page 10 hrs
5.6 Assignment management + grading annotation viewer 16 hrs
5.7 Analytics dashboard 10 hrs
5.8 Exercise center 8 hrs
5.9 Agent visual builder 12 hrs

5.10 Platform configuration page 4 hrs
5.11 Embeddable Widget 6 hrs

6.0 Platform Adaptation 16 hrs
6.1 Chaoxing adapter (simulated) + technical doc 8 hrs
6.2 DingTalk adapter (simulated) + technical doc 8 hrs

7.0 Testing & Quality Assurance 40 hrs
7.1 Unit tests (backend API, Agent logic) 12 hrs
7.2 Q&A accuracy validation (≥ 95%) 8 hrs
7.3 Grading accuracy validation (≥ 95%) 10 hrs
7.4 Performance test (500-user concurrency) 4 hrs
7.5 End-to-end integration testing 6 hrs

8.0 Documentation & Submission 32 hrs
8.1 Project overview introduction 4 hrs
8.2 Detailed technical solution document 12 hrs
8.3 Presentation PPT 8 hrs
8.4 Demo video recording & editing 8 hrs

WBS	**Module**	**Estimated Hours**	**Primary Responsible**
1.0	Project Management	24 hrs	All members
2.0	Requirements & System Design	40 hrs	All members
3.0	Infrastructure & DevOps	32 hrs	冯俊财 / 纪鹏
4.0	Backend Development	160 hrs	冯俊财 (AI/Agent) + 纪鹏 (API/data)
5.0	Frontend Development	96 hrs	张诗蔻
6.0	Platform Adaptation	16 hrs	纪鹏
7.0	Testing & QA	40 hrs	路清怡
8.0	Documentation & Submission	32 hrs	路清怡
	Total	**440 hrs**	

9. Project Budget Summary

Item	Description	Estimated Cost (CNY)
Cloud Server	4 vCPU / 8 GB RAM, rented as needed for deployment	¥ 100
LLM API Credits	Tongyi Qianwen (qwen-max / qwen-vl-max)	¥ 300
Object Storage	Aliyun OSS / MinIO for course files	¥ 50
Total		**¥ 450**

10. Quality Issues

Specific quality requirements:

Requirement	Metric	Target
Q&A accuracy	Core knowledge-point answer correctness	≥ 95%
Grading annotation accuracy	Key error annotation (location + content) correctness on sample assignments	≥ 95%
System availability	Uptime during demo and evaluation period	≥ 99%
API response time	Non-AI endpoint latency	< 2 seconds
Concurrency	Simultaneous active users	≥ 500

Quality assurance approach:

- All backend pull requests require peer code review before merging.
- Q&A accuracy is measured against a 50-question ground-truth set per course.
- Grading accuracy is measured against the sample assignments provided by the competition (key-error annotations pre-labeled by team members).
- Celery async tasks and Redis caching are used to maintain responsiveness under load.

11. Resources Required

People:

- 4 team members (roles defined in Section 2)

Technology:

Category	Tool / Technology
Frontend	React 18, TypeScript, Vite, Ant Design 5, ECharts, React Flow
Backend	FastAPI (Python 3.11), SQLAlchemy 2.0, Alembic, Celery
AI / LLM	LangChain 0.3, LangGraph, Tongyi Qianwen (qwen-max, qwen-vl-max), DeepSeek
Database	PostgreSQL 16 + pgvector, Redis 7, MinIO
DevOps	Docker Compose, Nginx, Aliyun ECS, GitHub Actions
Collaboration	GitHub (code), Feishu (communication), VS Code

Facilities:

- Cloud server (Aliyun ECS, provisioned by Week 2)
- Each member's local development machine
- GitHub repository for version control and CI

12. Assumptions and Risks

Assumptions:

1. The team has valid API access to Tongyi Qianwen (qwen-max, qwen-vl-max) throughout the project.
2. Competition rules allow simulated platform adaptation; real Chaoxing/DingTalk API certification is not required.
3. All team members can commit approximately 10 hours per week to the project.

Key Risks:

Risk	Probability	Impact	Mitigation Strategy
LLM API latency degrades user experience	Medium	High	SSE streaming output; Redis caching for frequent questions
Grading / Q&A accuracy fails to reach 95%	Medium	High	Per-type prompt templates; few-shot examples; iterative calibration with sample data
Schedule slippage due to exam season (late May)	Medium	Medium	Front-load core AI features in Weeks 3–7; documentation and video in final 2 weeks
Insufficient demo data for convincing showcase	Low	Medium	路清怡 dedicates time in Weeks 1–4 to prepare 2–3 complete course datasets
Chaoxing/DingTalk real integration complexity	Low	Low	Competition rules permit technical-spec-only adaptation; no live integration required

Constraints:

- Fixed end date: June 14, 2026; no extensions possible.
- Budget capped at approximately ¥ 800 (cloud + LLM API costs).
- Team size is fixed at 4 members; no external contractors.

13. Project Administration

Communications Plan:

- Weekly team sync meeting (Sunday evenings, 60 min) to review progress against milestones.
- Daily async updates via group chat (Feishu) for blockers and code review requests.
- GitHub Issues used to track bugs; GitHub Projects board tracks WBS task status.
- All code merged to `main` branch via Pull Requests only.

Scope Management Plan:

- Any proposed scope changes must be evaluated for impact on the June 14 deadline before acceptance.
- Core features (Q&A, grading, analytics, exercises) take priority. Creative features (Agent visual builder, platform adapter) are secondary if time is short.
- The Project Manager (冯俊财) has final authority on scope decisions.

Quality Management Plan:

- Mandatory peer code review for all pull requests before merging.
- Q&A and grading accuracy tested at each milestone; results recorded and prompt templates revised if targets are missed.
- End-to-end integration test performed in Week 10 covering all 9 UI pages and all 8 API route groups.

14. Acceptance and Approval

This Project Charter is approved by all team members, confirming the project objectives, scope, schedule, and responsibilities described above.

Role	Name	Signature	Date
Team Leader / Project Manager	冯俊财		March 29, 2026
Team Member	张诗蔻		March 29, 2026
Team Member	纪鹏		March 29, 2026
Team Member	路清怡		March 29, 2026

15. Appendices

Appendix A — Gantt Chart

> W1 = Mar 30–Apr 5 … W11 = Jun 8–Jun 14 | ▌ active | ▒ testing/review

Task	W1 3/30	W2 4/6	W3 4/13	W4 4/20	W5 4/27	W6 5/4	W7 5/11	W8 5/18	W9 5/25	W10 6/1	W11 6/8
1.0 Project Management	■	■	■	■	■	■	■	■	■	■	■
2.0 Requirements & System Design	■	■									
3.0 Database Schema & API Spec		■	■								
3.0 Infrastructure & DevOps	■	■									■
4.1 Auth & Core Models		■	■								
4.2 Agent SDK + LLMProvider			■	■							
4.3 Knowledge Middleware				■	■						
4.4a RAG Q&A Chain					■	■					
4.4b Grading Engine (text + code)						■	■				
4.4c Multimodal Grading (images)							■				
4.4d Learning Analytics Engine							■	■			
4.4e Exercise Generation Engine								■			
4.5 API Routes (all 8 groups)				■	■	■	■	■			
4.6 LangGraph Workflow Engine							■	■			
5.1 Login / Dashboard			■	■							
5.2 Course Management page				■	■						
5.3 Q&A Chat page					■	■					
5.4 Assignment + Grading pages						■	■				
5.5 Analytics Dashboard							■	■			
5.6 Exercises + Agent Builder								■	■		
5.7 Embedded Widget									■		
6.0 Platform Adaptation (tech spec)								■	■		
7.0 Unit & Accuracy Testing	▒	▒	▒	▒	▒	▒	▒	■	■	■	
7.0 Integration & E2E Testing									■	■	
7.0 Performance Testing (500-user)										■	
8.0 Cloud Deployment										■	■
8.0 Documentation & PPT										■	■
8.0 Demo Video Recording											■

MILESTONES:	M1	M2	M3	M4	M5
	Apr 12	Apr 26	May 17	May 31	Jun 14